Q2: When Does Short-Circuit Evaluation Matter?

CODE:

```
#include <stdio.h>

int calls = 0;

int verify_card(int disability)
{
   calls++;
   return disability;
}

int main()
{
   int i;
   int senior, eligible, disability;
   int count1 = 0, count2 = 0;
   int count3 = 0, count4 = 0;

   // Experiment 1: Cheap conditions first
   calls = 0;

   for (i = 0; i < 1000; i++)
   {
      senior = (i % 4 == 0);
      eligible = (i % 3 != 0);
      disability = (i % 10 == 0);
```

```
   if ((senior && eligible) || verify_card(disability))
      count1++;
}

printf("Experiment 1 - Calls: %d\n", calls);

// Experiment 2: Slow condition first
calls = 0;

for (i = 0; i < 1000; i++)
{
   senior = (i % 4 == 0);
   eligible = (i % 3 != 0);
   disability = (i % 10 == 0);

   if (verify_card(disability) || (senior && eligible))
      count2++;
}

printf("Experiment 2 - Calls: %d\n", calls);

// Experiment 3: Bitwise operators
calls = 0;

for (i = 0; i < 1000; i++)
{
   senior = (i % 4 == 0);
```

```
    eligible = (i % 3 != 0);
    disability = (i % 10 == 0);

    if ((senior && eligible) | verify_card(disability))
        count3++;
}

printf("Experiment 3 - Calls: %d\n", calls);

// Experiment 4: Nested if statements
calls = 0;

for (i = 0; i < 1000; i++)
{
    senior = (i % 4 == 0);
    eligible = (i % 3 != 0);
    disability = (i % 10 == 0);

    if (senior && eligible)
        count4++;
    else if (verify_card(disability))
        count4++;
}

printf("Experiment 4 - Calls: %d\n", calls);

// Changed condition: Audit every passenger
calls = 0;
```

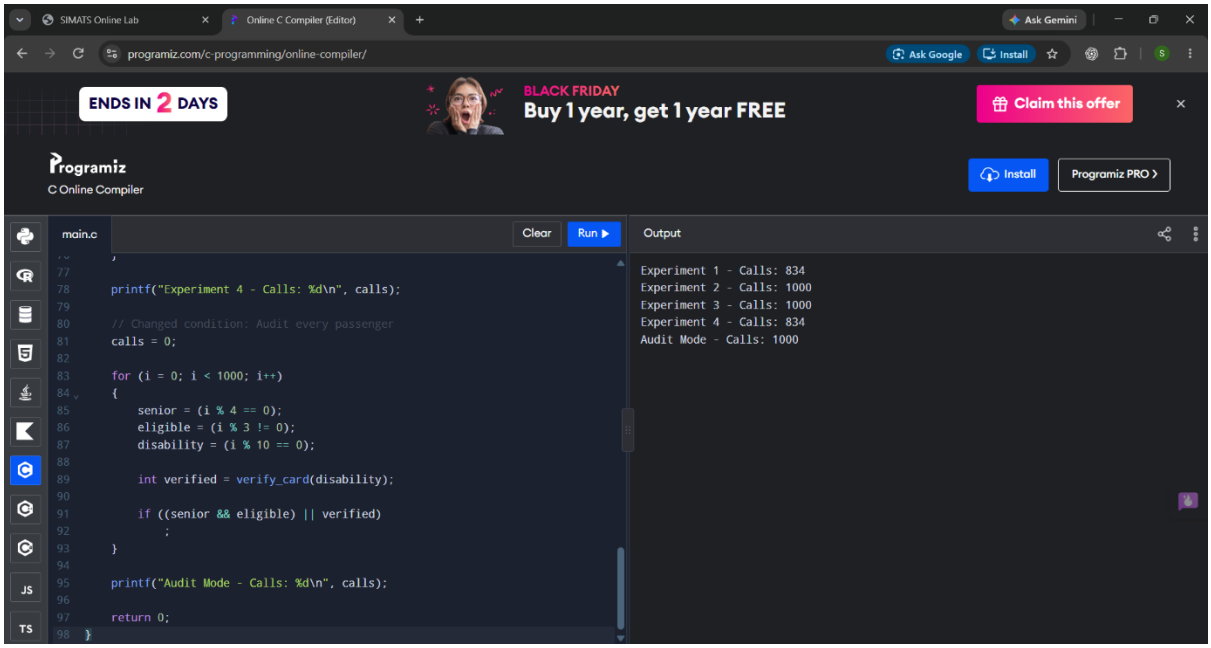
```
    for (i = 0; i < 1000; i++)
    {
        senior = (i % 4 == 0);
        eligible = (i % 3 != 0);
        disability = (i % 10 == 0);

        int verified = verify_card(disability);

        if ((senior && eligible) || verified)
            ;
    }

    printf("Audit Mode - Calls: %d\n", calls);

    return 0;
}
```


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Programiz PRO
main.c
Clear
Run
Output
printf("Experiment 4 - Calls: %d\n", calls);
// Changed condition: Audit every passenger
calls = 0;
for (i = 0; i < 1000; i++)
{
senior = (i % 4 == 0);
eligible = (i % 3 != 0);
disability = (i % 10 == 0);
int verified = verify_card(disability);
if ((senior && eligible) || verified)
;
}
printf("Audit Mode - Calls: %d\n", calls);
return 0;
}
Experiment 1 - Calls: 834
Experiment 2 - Calls: 1000
Experiment 3 - Calls: 1000
Experiment 4 - Calls: 834
Audit Mode - Calls: 1000